JESTER N. ITLIONG

Computational Physicist • Physics Educator • Scientific Software Developer

Ph.D. Candidate in Physics, Michigan Technological University (defense: July 27, 2026)
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PROFESSIONAL SUMMARY

Ph.D. candidate in Physics with expertise in multiscale molecular simulations of polymers, ionic liquids, and soft materials. Experienced in atomistic and coarse-grained molecular dynamics, statistical mechanics, and high-performance scientific computing to investigate transport, rheological, dielectric, and structural properties of complex polymeric systems. Over 8 years of interdisciplinary computational research experience and over 4 years of university teaching, complemented by open-source scientific software including machine-learning tools and interactive simulations. Research connects molecular interactions, mesoscale organization, and macroscopic material behavior through scalable computational workflows and physics-based modeling.

RESEARCH INTERESTS

- Multiscale modeling of polymers and soft materials
- Coarse-grained molecular dynamics
- Structure–property relationships
- Polymer transport phenomena
- Rheology and relaxation dynamics
- Mesoscale organization in polymeric systems
- Statistical mechanics of soft matter
- Machine learning for materials properties

EDUCATION

Michigan Technological University — Ph.D. in Physics *Jan 2021 – present*

Houghton, MI, USA • GPA: 4.00/4.00 • Advisor: Prof. Issei Nakamura

- Dissertation: *Molecular Origins of Viscosity in Ionic and Polymerized Ionic Liquids: Insights from Stockmayer Fluid Simulations* (defense scheduled July 27, 2026)
- Graduate Certificate in Advanced Computational Physics (2023)

De La Salle University — M.S. in Physics *2017*

Manila, Philippines • Advisor: Prof. Nelson B. Arboleda, Jr.

- Thesis: atomistic MD of water transport and reverse solute diffusion across polyamide membranes in forward-osmosis dewatering of microalgae

Eulogio “Amang” Rodriguez Institute of Science and Technology — B.S. in Applied Physics (minor: Computer Science) *2014*

Manila, Philippines • Cum Laude

RESEARCH EXPERIENCE

Graduate Research Assistant — Michigan Technological University *Jan 2021 – present*

Computational soft matter physics • Collaboration with Sandia National Laboratories

- Develop coarse-grained MD models of ionic liquids and polymerized ionic liquids to investigate viscosity, ionic conductivity, dielectric behavior, and chain dynamics
- Perform equilibrium and non-equilibrium LAMMPS simulations; analyze transport with Green–Kubo, Einstein–Helfand, and correlation-function methods
- Design scalable Python/Slurm workflows for automated simulation analysis and large-scale statistical-mechanical calculations on HPC resources
- Validate coarse-grained predictions against experimental viscosity and glass-transition data
- Mentor undergraduate and graduate researchers in molecular simulation, scientific programming, and technical communication

Research Assistant — De La Salle University *2018 – 2020*

Manila, Philippines

- Performed atomistic MD of polymer, membrane, and electrolyte systems (GROMACS) to investigate water and solute transport in soft materials
- Developed force-field parameters for polymer systems using density functional theory

TEACHING EXPERIENCE

Assistant Professorial Lecturer (Part-time) — De La Salle University *2018 – 2020*

Physics, Optics, Electronics, and Modern Physics Laboratories (100+ students/year)

- Developed active-learning laboratory activities and assessments aligned with course outcomes

Instructor (Full-time) — EARIST *2017 – 2020*

Mechanics; Electromagnetism; Statistical Physics; Quantum Physics; Computational Physics (500+ students)

- Integrated Python programming and computational methods into advanced physics instruction
- Supervised undergraduate research projects from design through presentation

K–12 Science Teacher and College Instructor — De Ocampo Memorial College *2014 – 2015*

Biology, chemistry, and physics; introductory courses for pre-medical undergraduates

TEACHING EFFECTIVENESS AND MENTORING

Evaluations	Undergraduate Physics Laboratory: 4.72/5.00 (AY 2018–19) and 4.88/5.00 (AY 2019–20), both rated Outstanding
Student feedback	“Teaches with clarity and depth.” • “Turns complicated topics understandable.” • “Very approachable and teaches amazingly.”
Mentoring	Mentored undergraduate and graduate students in molecular simulation, scientific programming, data analysis, technical writing, and presentations; supervised research and capstone projects

SOFTWARE AND OPEN-SOURCE PROJECTS

Ionic-Liquid Property Explorer *2026*

il-property-explorer.streamlit.app • github.com/jnitlion/il-property-explorer

- Machine-learning dashboard predicting viscosity, ionic conductivity, density, and glass-transition temperature of ionic liquids from molecular structure, trained on 65,000+ experimental measurements from the NIST ILThermo database
- End-to-end pipeline: automated data harvesting, curation, VFT curve fitting, RDKit per-ion featurization, and quantile gradient-boosting models validated on held-out ion families

Interactive Stockmayer Fluid Simulator *2026*

jnitlion.github.io/projects/stockmayer-simulator.html

- Browser-based 3D molecular dynamics simulation of dipolar fluids (velocity-Verlet integration, thermostat/barostat, dipole–dipole interactions) for teaching and outreach; plain JavaScript + WebGL

Physics Browser Games: Maxwell’s Demon and Don’t Get Glassy *2026*

jnitlion.github.io/projects/maxwells-demon.html • jnitlion.github.io/projects/dont-get-glassy.html

- Educational physics games built on real molecular simulation: sorting a two-chamber gas against the Second Law with a live mixing-entropy meter, and fighting kinetic arrest in a bidisperse liquid under a Langevin quench toward the glass transition
- Interactive outreach companions to my dissertation research on glassy dynamics; plain JavaScript + HTML canvas, no frameworks

PUBLICATIONS

[6] **Itliong, J.N.**, Frischknecht, A.L., Stevens, M.J., and Nakamura, I. (2026). Viscosity, Chain Dynamics, and Ion-pairing in Polymerized Ionic Liquids: Effects of Cation Size and Dipole Mobility. *Submitted to Macromolecules.*

[5] **Itliong, J.N.**, Frischknecht, A.L., Stevens, M.J., and Nakamura, I. (2025). Stockmayer Fluid Simulations for Viscosity and Glass Transition Temperature of Ionic Liquids. *The Journal of Chemical Physics*, 163(4), 044503. doi:10.1063/5.0268727

[4] Gao, T., **Itliong, J.**, Kumar, S.P., Hjorth, Z., and Nakamura, I. (2021). Polarization of Ionic Liquid and Polymer and its Implications for Polymerized Ionic Liquids: An Overview Towards a New Theory and Simulation. *Journal of Polymer Science*, 59(21), 2434–2457. doi:10.1002/pol.20210330

[3] **Itliong, J.N.**, Villagracia, A.R.C., Moreno, J.L.V., et al. (2019). Investigation of reverse ionic diffusion in forward-osmosis-aided dewatering of microalgae: A molecular dynamics study. *Bioresource Technology*, 279, 181–188. doi:10.1016/j.biortech.2019.01.109

[2] **Itliong, J.N.**, Villagracia, A.R.C., Ong, H.L., et al. (2019). H_2O Absorptivity on a Fully 4-crosslinked Polyacrylamide Membrane via Density Functional Theory and Monte Carlo Calculations for Draw Solution Recovery in Forward Osmosis. *2019 IEEE 11th HNICEM*, Laoag, Philippines. doi:10.1109/HNICEM48295.2019.9072782

[1] **Itliong, J.N.**, Villagracia, A.R.C., Moreno, J.L.V., et al. (2018). Density Functional Theory-based modeling and calculations of a polyamide molecular unit for studying forward-osmosis-dewatering of microalgae. *2018 IEEE 10th HNICEM*, Baguio City, Philippines. doi:10.1109/HNICEM.2018.8666436

SELECTED PRESENTATIONS

"Role of Cation Size and Dipolar Motion Restriction in the Viscosity of Polymerized Ionic Liquids." *APS Global Physics Summit, Denver, CO*, Mar 2026.

"Effects of Molecular Size and Polarity on Viscosity Scaling and Ionic Conductivity of Polymerized Ionic Liquids." *APS Global Physics Summit, San Diego, CA*, Mar 2025.

"Influences of Chain Polarity and Molecular Weight on Ion and Polymer Dynamics in Polymerized Ionic Liquids." *APS March Meeting, Minneapolis, MN*, Mar 2024.

"Effects of Electric Charge, Dipole Moment, and Molecular Size on the Viscosity of Ionic Liquids." *APS March Meeting, Las Vegas, NV*, Mar 2023.

Resource Speaker, APS Conference for Undergraduate Women and Gender Minorities in Physics. *Michigan Technological University*, Jan 2025.

HONORS AND AWARDS

Finalist, Midwest Regional 3-Minute Thesis Competition *2026*

Midwestern Association of Graduate Schools, Kansas City, MO

Recipient, Finishing Fellowship Award for Summer 2026 *2026*

Graduate School, Michigan Tech

1st Place, 3-Minute Thesis Competition *2025*

Graduate Student Government, Michigan Tech (13 finalists)

Winner, T-Shirt Design Competition *2025*

APS Division of Polymer Physics

Best Poster Presentation *2023*

Graduate Physics Colloquium, Michigan Tech

Outstanding Faculty Award *2019*

College of Arts and Sciences, EARIST (voted by 100+ students)

LEADERSHIP AND SERVICE

2023 – 2024	**Graduate Physics Representative**, Graduate Student Government, Michigan Tech — advocated for graduate student welfare; developed a peer-evaluation procedure for the Graduate Physics Colloquium
2025	**Volunteer**, APS Division of Polymer Physics — membership and conference operations, APS March Meeting 2025
2018 – 2020	**Research Coordinator**, College of Arts and Sciences, EARIST — organized institution-wide research colloquia; developed a systematic archiving workflow for student research

PROFESSIONAL MEMBERSHIPS

American Physical Society, Division of Polymer Physics (2021 – present) • Women in Physics at Michigan Tech (2021 – present) • Philippine Association of Physics and Science Instructors (2015 – 2020)

TECHNICAL SKILLS

Simulation & modeling	Molecular dynamics, coarse-grained modeling, statistical mechanics, transport phenomena, rheology and viscosity calculations, Monte Carlo methods
Scientific computing	Python, C++, Fortran, MPI/OpenMP, Slurm, Linux, high-performance computing
Simulation software	LAMMPS, GROMACS, OVITO, VMD, Gaussian, Avogadro
Data science & ML	NumPy, SciPy, pandas, scikit-learn, RDKit, matplotlib, Streamlit; gradient-boosting and Gaussian-process models with uncertainty quantification
Teaching technologies	Canvas, Zoom, Microsoft Office, Google Workspace

REFERENCES

Issei Nakamura
Associate Professor of Physics
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